



Subject: Computer Networks

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Student Name: Trimann Kaur

Submitted to: Mr. Alok

UID: 24BAI70511

Branch: B.E. CSE (AIML)

Section/Group: 24AIT_KRG-1-B

Assignment-2

Q1. What is an IP Address? Why do we need classes in case of IPV4 address? Explain each class. Mention how many host and networks are possible in each class. Also explain what is subnetting and supernetting with examples.

Ans –

IP Address

An IP address is a logical address given to a device connected to a network. It is used to identify the device and helps in sending data from one device to another.

IPv4 uses a 32-bit address, which is divided into four parts of 8 bits each. Each part of an IPv4 address can have a value from 0 to 255.

For example – 192.168.1.1

Need for Classes in IPv4

In the earlier IPv4 addressing system, IP addresses were divided into different classes. This was done because different networks had different requirements.

For example, a very large organization may require thousands of host addresses, whereas a small organization may require only a few hundred. Therefore, different classes were made for different network sizes. Each class provides a different number of bits for the Network ID and the Host ID.

The main IPv4 classes are Class A, B, C, D and E.

Class A

Class A is mainly used for very large networks.

- First octet range: 1 to 126
- Default subnet mask: 255.0.0.0
- Network bits: 8
- Host bits: 24
- Number of networks: $2^7 - 2 = 126$
- Number of usable hosts per network: $2^{24} - 2 = 16,777,214$

Example – 10.0.0.1

Here, the first bit is fixed as 0, so 7 bits are available for the network number. Two network values are excluded, giving 126 networks.

Class B

Class B is used for medium-sized networks.

- First octet range: 128 to 191
- Default subnet mask: 255.255.0.0
- Network bits: 16
- Host bits: 16
- Number of networks: $2^{14} = 16,384$
- Number of usable hosts per network: $2^{16} - 2 = 65,534$

Example – 172.16.0.1

Here, the first two bits are fixed as 10, so 14 bits are available for identifying networks.

Class C

Class C is generally used for small networks.

- First octet range: 192 to 223
- Default subnet mask: 255.255.255.0
- Network bits: 24
- Host bits: 8
- Number of networks: $2^{21} = 2,097,152$
- Number of usable hosts per network: $2^8 - 2 = 254$

Example – 192.168.1.1

Here, the first three bits are fixed as 110, so 21 bits are available for identifying networks.

Class D

Class D addresses range from 224.0.0.0 to 239.255.255.255.

- The first four bits are fixed as 1110.
- These addresses are mainly used for multicasting.
- It does not have separate network and host portions like Classes A, B and C.

Class E

Class E addresses range from 240.0.0.0 to 255.255.255.255.

- The first four bits are fixed as 1111.
- The addresses are reserved mainly for experimental and special purposes.
- It does not have normal network and host divisions.

Class	First Octet Range	Default Mask	Number of Networks	Hosts per Network
A	1–126	255.0.0.0	126	16,777,214
B	128–191	255.255.0.0	16,384	65,534
C	192–223	255.255.255.0	2,097,152	254
D	224–239	Not applicable	Multicast	Not applicable
E	240–255	Not applicable	Reserved	Not applicable

The number of usable hosts is calculated using: $2^{(\text{number of host bits})} - 2$

Two addresses are excluded because one is used as the network address and the other is used as the broadcast address.

Subnetting

Subnetting means dividing one large network into smaller networks called subnets. It is useful because it:

- Makes better use of available IP addresses
- Reduces unnecessary network traffic
- Makes network management easier
- Provides better separation between different parts of a network

For example, consider the network:

192.168.1.0/24

A /24 network has 24 network bits, so the remaining:

$32 - 24 = 8$ bits

are available for hosts.

If we divide this network using /26, then:

$$26 - 24 = 2 \text{ bits}$$

are borrowed from the host portion for subnetting.

Number of subnets:

$$2^2 = 4 \text{ subnets}$$

because 2 bits are borrowed for subnetting.

After borrowing 2 bits, the remaining host bits are:

$$8 - 2 = 6 \text{ host bits}$$

Therefore, the total number of addresses in each subnet is:

$$2^6 = 64 \text{ addresses}$$

because 6 bits are available for hosts.

Out of these 64 addresses, one is reserved as the network address and one as the broadcast address.

Therefore, $64 - 2 = 62$ usable hosts per subnet

Hence, the /24 network is divided into 4 subnets, with 64 addresses and 62 usable host addresses in each subnet.

Supernetting

Supernetting is the opposite of subnetting. In supernetting, multiple smaller networks are combined to form one larger network. It is mainly used to reduce the number of entries in routing tables.

For example, consider these networks:

- 192.168.0.0/24
- 192.168.1.0/24
- 192.168.2.0/24
- 192.168.3.0/24

These four networks can be combined as: 192.168.0.0/22

Instead of maintaining four different routing entries, a router can maintain only one combined entry.

Q2. What is the difference between Link state and Distance vector routing? Explain how these two algorithms help in identifying the least cost path with the help of examples.

Ans –

Routing algorithms are used by routers to find a suitable path for sending packets from source to destination. Two common routing methods are Distance Vector Routing and Link State Routing.

Distance Vector

- It knows the distance to a destination through neighbouring routers.
- It shares its routing table with neighbouring routers.
- It mainly uses the Bellman-Ford Algorithm.
- It generally has slower convergence.
- It has a higher chance of routing loops.
- It is comparatively less complex.
- It requires fewer system resources.
- RIP is an example of a Distance Vector routing protocol.

Link State

- It knows the complete network topology.
- It shares information about its connected links with other routers.
- It mainly uses Dijkstra's Algorithm.
- It generally has faster convergence.
- It has a lower chance of routing loops.
- It is comparatively more complex.
- It requires more system resources.
- OSPF is an example of a Link State routing protocol.

Distance Vector Routing

In Distance Vector Routing, each router maintains information about the destination, cost and next hop. It exchanges this information with neighbouring routers and uses the Bellman-Ford Algorithm to find the least-cost route.

Example –

If

A → B = 1

B → C = 2

A → C = 5

Then the cost through B is $1 + 2 = 3$

Since $3 < 5$, the least-cost path is $A \rightarrow B \rightarrow C$

Link State Routing

In Link State Routing, every router gets information about the network links and builds a complete map of the network. It then uses Dijkstra's Algorithm to find the shortest path.

Example:

Suppose

- A to B = 2
- B to C = 1
- C to D = 2
- B to D = 5

Then:

$A \rightarrow B \rightarrow D = 2 + 5 = 7$

$A \rightarrow B \rightarrow C \rightarrow D = 2 + 1 + 2 = 5$

Therefore, the least-cost path is $A \rightarrow B \rightarrow C \rightarrow D$ with a cost of 5.

Thus, Distance Vector depends mainly on information from neighbouring routers, while Link State uses the complete network topology.

Q3. What is congestion in a network? Why do we need to control congestion? Explain the difference between Leaky bucket and Token bucket with examples. Also, explain why do we need to do fragmentation?

Ans –

Network congestion

Network congestion occurs when the amount of traffic entering a network becomes greater than the network's capacity to handle it. When too many packets arrive at routers or communication links at the same time, packets start waiting in queues. For example, if a router can process 1000 packets per second but receives 2000 packets per second, the extra packets have to wait. If the buffer becomes full, some packets may be dropped.

Congestion control is needed to prevent the network from becoming overloaded.

It helps in:

- Reducing packet loss
- Reducing network delay
- Avoiding buffer overflow
- Improving network performance
- Making efficient use of available bandwidth
- Preventing unnecessary retransmissions

Two common traffic-control techniques are the Leaky Bucket Algorithm and the Token Bucket Algorithm.

Leaky Bucket Algorithm

The Leaky Bucket Algorithm sends packets at a fixed rate, even if they arrive at different rates. Extra packets may be dropped if the bucket becomes full.

Example – If the bucket can hold 10 packets and sends 2 packets per second, then 8 packets arriving together are stored and transmitted at 2 packets per second.

Token Bucket Algorithm

In the Token Bucket Algorithm, tokens are generated at a fixed rate. A packet can be sent only when a token is available. Stored tokens allow packets to be sent in bursts.

Example – If 2 tokens are generated per second, then after 5 seconds, $2 \times 5 = 10$ tokens
If 10 packets arrive, they can be sent using these stored tokens.

Leaky Bucket

- It sends packets at a mostly fixed rate.
- It does not handle burst traffic efficiently.
- It stores packets in the bucket.
- Extra packets may be dropped when the bucket is full.
- It produces a smooth and regular traffic flow.
- It is mainly used for traffic shaping.

Token Bucket

- It can send packets at a variable rate.
- It allows controlled burst traffic.
- It stores tokens in the bucket.
- Extra tokens are discarded when the bucket is full.
- It provides a more flexible traffic flow.
- It is mainly used for rate control and burst handling.

Therefore, Leaky Bucket is suitable when a fixed and smooth output rate is required, while Token Bucket is more suitable when temporary bursts of traffic are allowed.

Fragmentation

Fragmentation is the process of dividing a large IP packet into smaller pieces called fragments. It is required because different networks may support different maximum packet sizes. The maximum size of a packet that can be carried by a network is called the Maximum Transmission Unit (MTU).

For example, suppose an IP packet has a size of 4000 bytes but the next network has an MTU of 1500 bytes. The 4000-byte packet cannot be transmitted directly through that network.

Therefore, in IPv4, it may be divided into smaller fragments that can fit within the MTU limit.

These fragments travel separately through the network and are reassembled at the destination.

Fragmentation is needed because:

- Different networks can have different MTU values.
- A packet may be larger than the size supported by the next network.
- It allows large packets to travel through networks with smaller frame-size limits.
- It makes communication possible between networks using different technologies.

Hence, fragmentation allows packets to travel successfully through networks having different MTU limitations.